

Question: What is the ample cone of $\mathcal{L}_Y(\lambda)$?

We begin by reviewing the history of this problem:

1. In [2], the authors compute the ample cone for Hilbert modular surface, by constructing partial Hasse invariants and studying intersection theory on the surface directly.
2. In [40], the authors conjectured the ampleness criterion for $U(2)$ /Hilbert/quaternionic Shimura varieties. They studied the geometric description of Goren–Oort strata and proved the necessity part of the criterion.
3. In [45], the author proved the ampleness criterion conjectured by [40], by computing the intersection numbers using the description of strata.
4. In the unpublished paper [5], the authors proposed a sufficiency criterion for ampleness for Hodge type Shimura varieties. However, their bound is known to be not sharp.

We know very little about the ampleness criterion beyond the above cases, especially when the group G is not type A_1 . The main difficulty is that the stratification of the flag space is usually very complicated, and computing intersection numbers $(\mathcal{L}_Y(\lambda) \cdot C)$ with a complete curve $C \subseteq Y$ is non-trivial.

In this paper, we give a sharp answer to the above question when the group G is of type A and $G_{\mathbb{Q}_p}$ is split unitary.

Statement of the main result.

Let F be a totally real field of degree N over $\mathbb{Q}$ in which p is inert. We label the set of real embeddings of F by $\{\tau_1, \dots, \tau_N\}$ as before. Let E/F be an imaginary quadratic extension in which $p\mathcal{O}_E = \mathfrak{q}\mathfrak{q}^c$. Each τ_i extends to complex embeddings τ_i (over $\mathfrak{q}$) and τ_i^c (over $\mathfrak{q}^c$). For a tuple of integers $(m_i, n_i)_{1 \leq i \leq N}$ such that $m_i + n_i = n$, let G be the unitary group over $\mathbb{Q}$, with signature at infinite places given by $(m_i, n_i)_{1 \leq i \leq N}$, i.e.,

$$G(\mathbb{R}) = G\left(\prod_{i=1}^N U(m_i, n_i)\right). \quad (7)$$

In Section 3, we realize G as the group of symplectic similitudes of an n -dimensional E -vector space. Let X be the special fiber of the corresponding Shimura variety of signature (m_i, n_i) . Here we apologize for using the letter X again (but for different meaning) to maintain consistency with the rest of the paper. The signature condition implies that in the Hodge filtration,

$$0 \longrightarrow \omega_{\mathcal{A}^\vee/X, \tau_i} \longrightarrow H_1^{\text{dR}}(\mathcal{A}/X)_{\tau_i} \longrightarrow \text{Lie}_{\mathcal{A}/X, \tau_i} \longrightarrow 0, \quad (8)$$

$\omega_{\mathcal{A}^\vee/X, \tau_i}$ (resp. $\omega_{\mathcal{A}^\vee/X, \tau_i^c}$) is a subbundle of $\omega_{\mathcal{A}^\vee/X}$ of rank m_i (resp. n_i). The polarization in the moduli problem induces a pairing

$$\langle \cdot, \cdot \rangle : H_1^{\text{dR}}(\mathcal{A}/X)_{\tau_i} \times H_1^{\text{dR}}(\mathcal{A}/X)_{\tau_i^c} \longrightarrow \mathcal{O}_X \quad (9)$$

such that $\omega_{\mathcal{A}^\vee/X, \tau_i^c} = \omega_{\mathcal{A}^\vee/X, \tau_i}^\perp$ under this pairing.